

Digitale tweelingen

Predictieve simulaties van wandelen voor de klinische gangbeeldanalyse

Tom Buurke, Friedl De Groote, Tim van der Zee, Ellis Van Can, Ines Vandekerckhove
Workshop SMALLL Congres 2025, Erasmus MC, Rotterdam, NL

umcg:

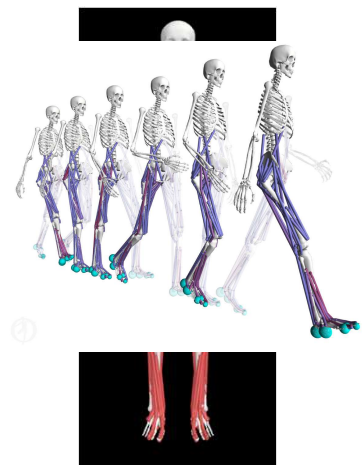


KU LEUVEN

1

Predictieve simulaties van wandelen voor de klinische gangbeeldanalyse

- Wat is een musculoskeletaal model?
 - Wiskundig model
 - Starre lichamen (botten)
 - Verbonden met gewrichten
 - Spieren over de gewrichten
 - Spieren genereren kracht en beweging



Workshop SMALLL 2025
Digitale tweelingen

2

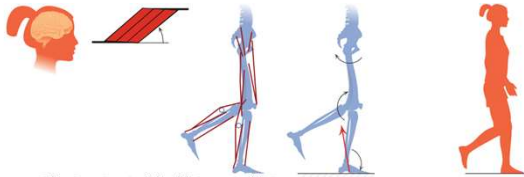
2

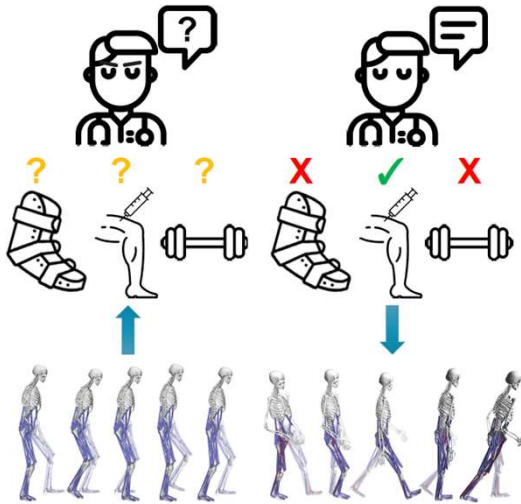
Waarom gebruiken we musculoskeletaal modellen?

Vanuit gemeten data

- Modelleren wat we niet kunnen meten
 - Spierkracht
 - Interne belasting
 - Spier-pees lengte

De beweging voorspellen

- 
- 



Illustrations from Uchida & Delp (2020)

Workshop SMALLL 2025
Digitale tweelingen

3

3

Hoe gebruiken we musculoskeletaal modellen?

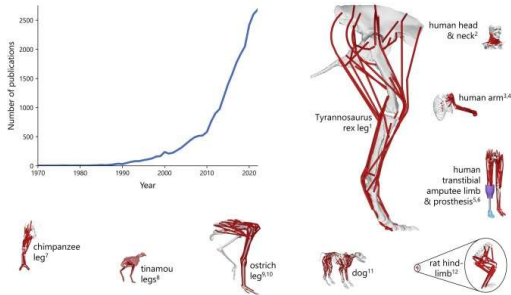
• Open-source software

- OpenSim
- PredSim
- SCONE
- Moco
- MyoSuite
- ...



• Simulation CPU time

- 2001: 10,000 hours¹
- 2019: 1 hour²



1) Anderson & Pandy (2001)
2) Falisse et al. (2019)

Workshop SMALLL 2025
Digitale tweelingen

Figure from Uhlrich et al. (2023)

4

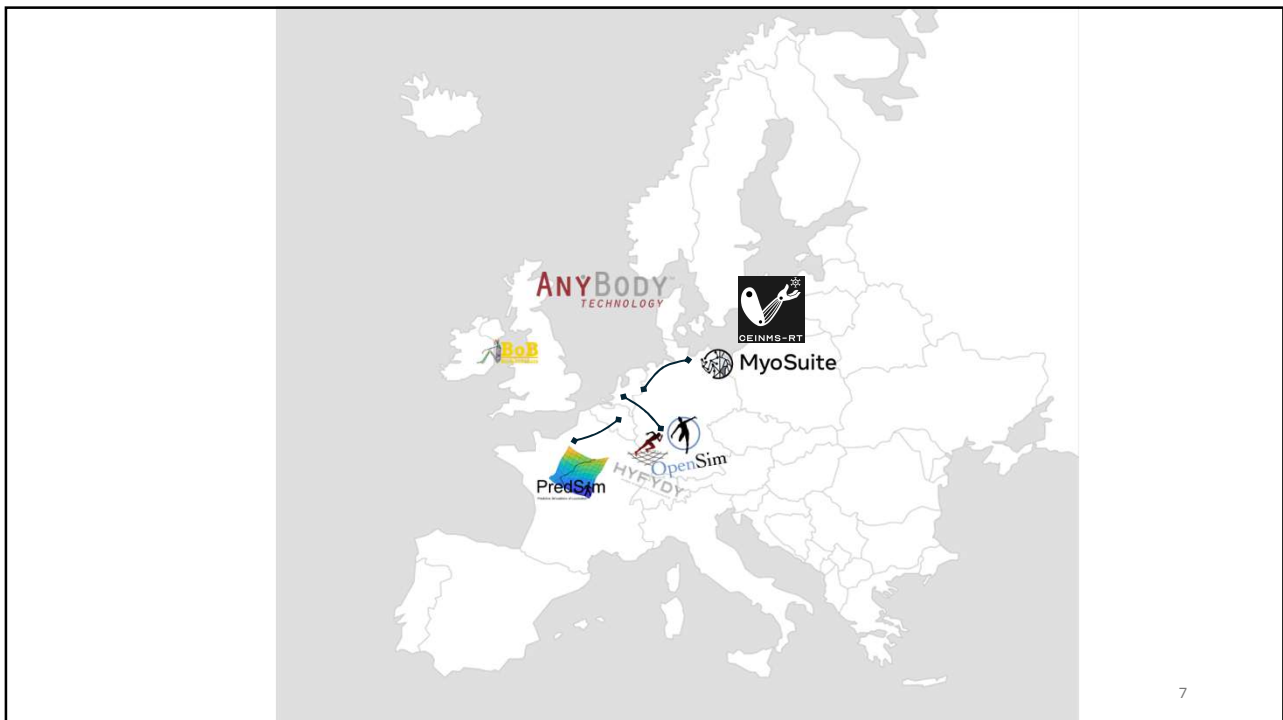
4



5



6



7

Het doel van vandaag

Na deze workshop...

- kent u de onderliggende theorie van musculoskeetaal modelleren en predictieve simulaties.
- kunt u predictieve simulaties met 2D modellen in *PredSim* uitvoeren.
- kunt u de informatie en bronnen voor 3D simulaties vinden.
- kent u kansen en obstakels voor de klinische implementatie van predictieve simulaties.



8

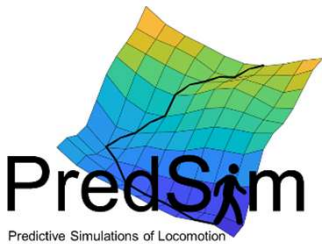
Programma

- 00:00 – 00:15: Introductie en theorie
- Predictieve simulaties voor de klinische gangbeeldanalyse (Tom Buurke)
 - Onderliggende theorie van modelleren en simulaties met *PredSim* (Friedl De Grootte)

- 00:15 – 00:30: De casussen van vandaag: Wat kunnen we personaliseren en simuleren?
- Casus 1: Spierzwakte en -stijfheid bij Duchenne spierdystrofie (Ines Vandekerckhove)
 - Casus 2: Chirurgische ingreep bij cerebrale parese (Ellis Van Can)
 - Casus 3: Het effect van een enkel-voet-orthese bij een dropvoet (Tim van der Zee)

00:30 – 01:00: Hands-on: Predictieve simulaties in 2D met *PredSim*

- 01:00 – 01:15: Afsluitende discussie
- Showcase: Predictieve simulaties in 3D (Friedl De Grootte)
 - Groepsdiscussie: Kansen en obstakels voor de klinische implementatie van predictieve simulaties (Tom Buurke)



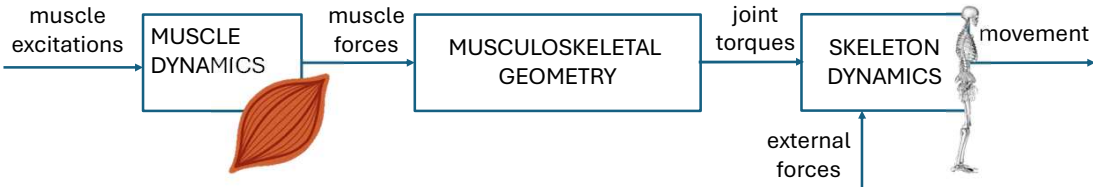
9

Simulations based on a musculoskeletal model



Monty Python: Ministry of silly walks

dynamic constraints

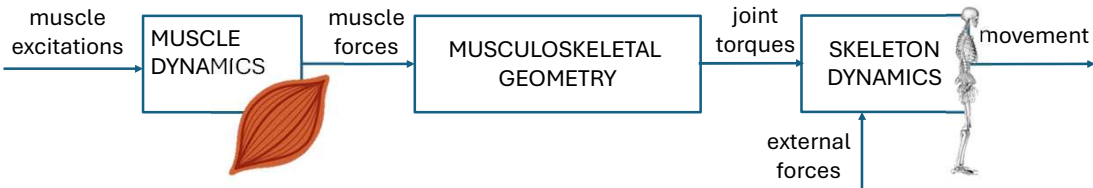


task constraints, e.g. average forward speed and periodicity for locomotion

10

Optimal control assumption

find muscle excitations **that**
minimize cost
subject to
dynamic constraints



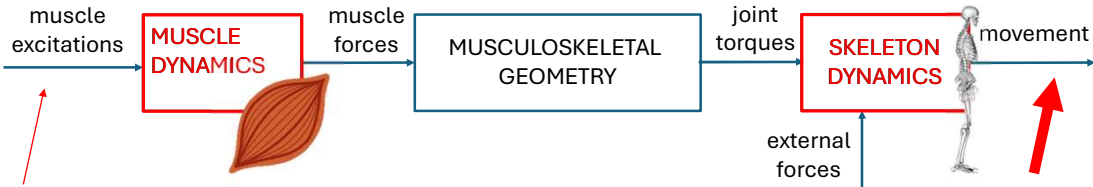
task constraints, e.g. average forward speed and periodicity for locomotion

11

Solving optimal control problems

find muscle excitations **that**
minimize cost
subject to
dynamic constraints

STIFF DIFFERENTIAL EQUATIONS



task constraints, e.g. average forward speed and periodicity for locomotion

12

We reduced computational cost by factor 20

Challenge: High computational cost of numerical methods

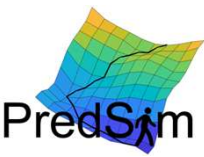
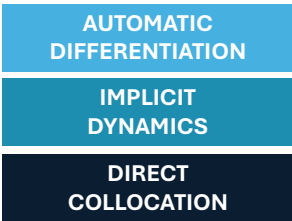
Muscle-driven 3D model (23 dofs, 54 muscles)
~10,000 hours (Anderson and Pandy, 2001)

Muscle-driven 3D model (23 dofs, 40 muscles)
~60 hours (Miller, 2014)

Muscle-driven 3D model (25 dofs, 80 muscles)
~13-17 hours (Lin ... Pandy, 2018)

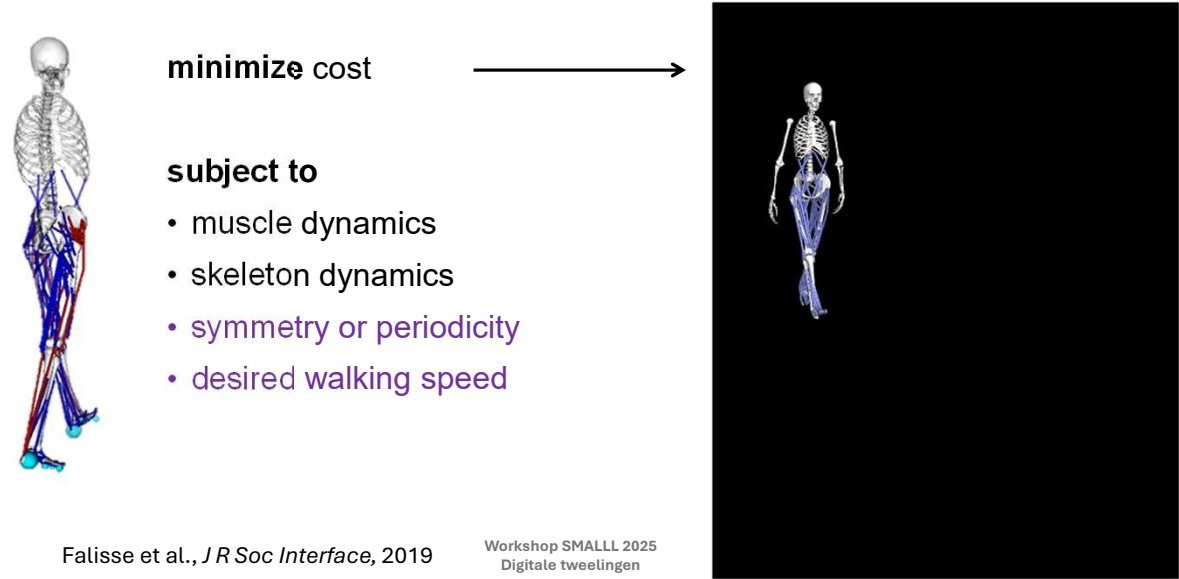
Muscle-driven 3D model (29 dofs, 92 muscles)
<1 hour (Falisse ... De Groote, 2019)

2001
2014
2018
2019

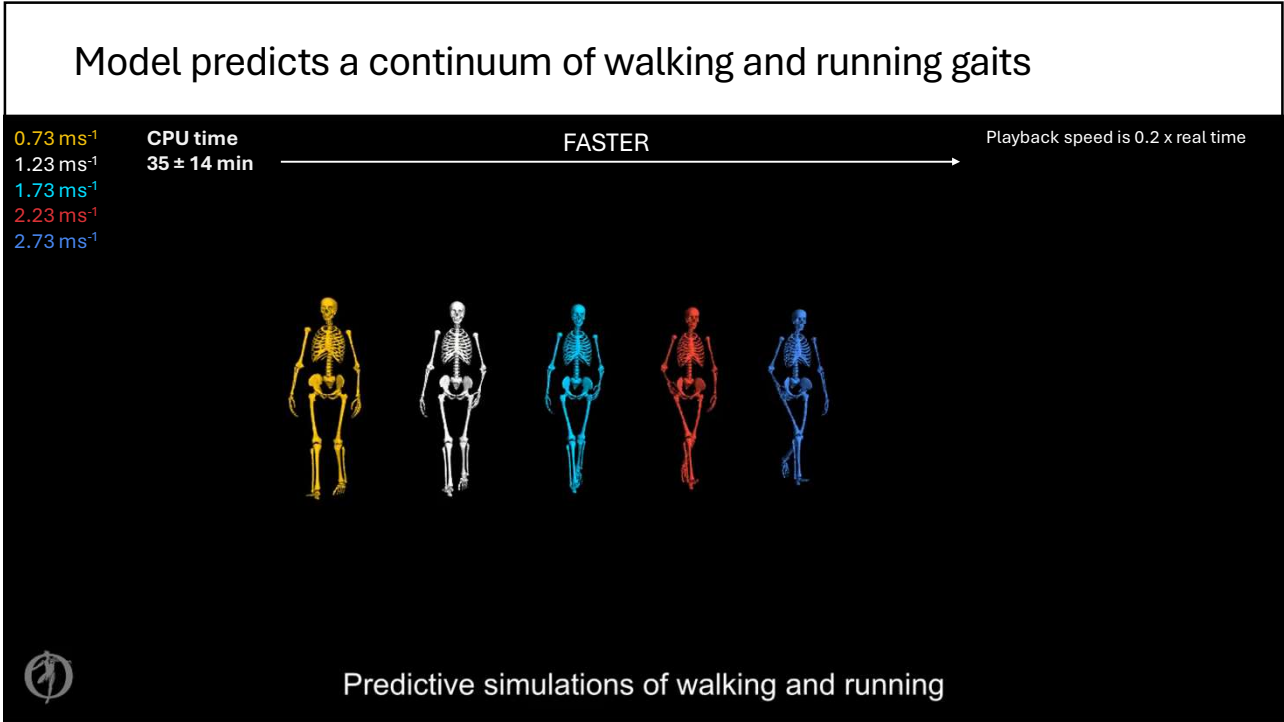


13

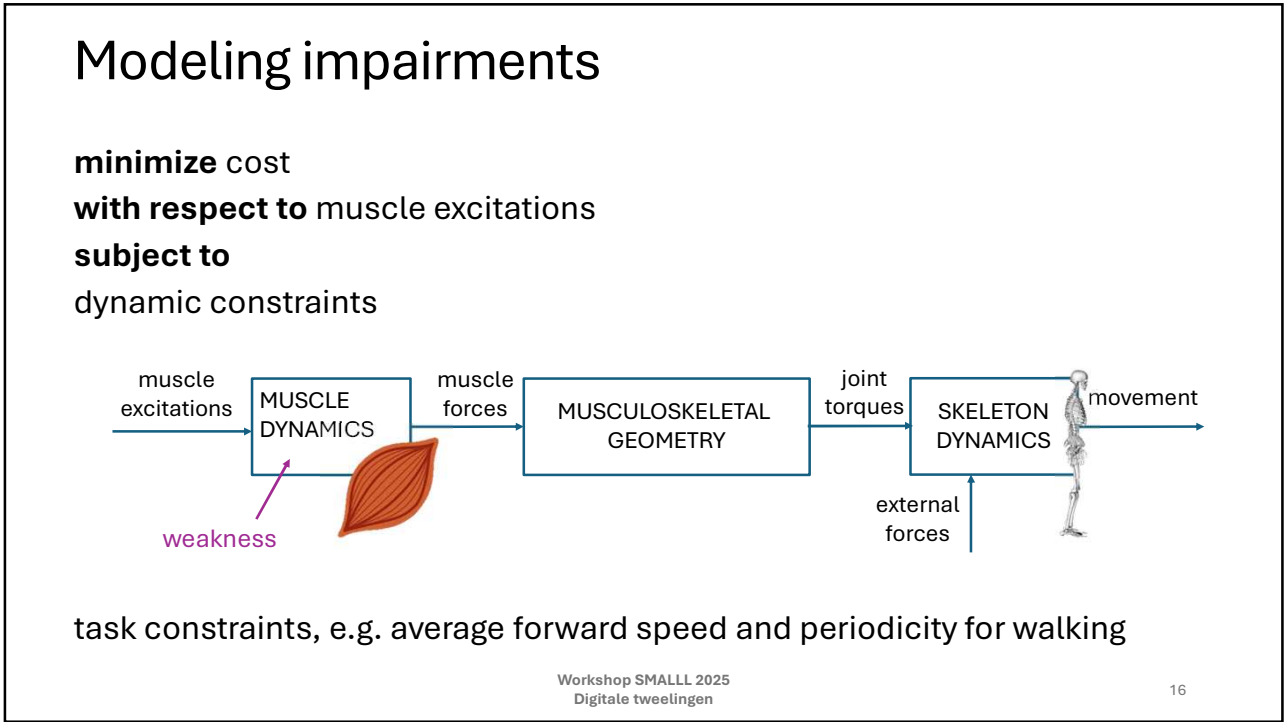
Predictive simulations of muscle driven 3D walking allow us to explore effect of neuro-musculoskeletal deficits



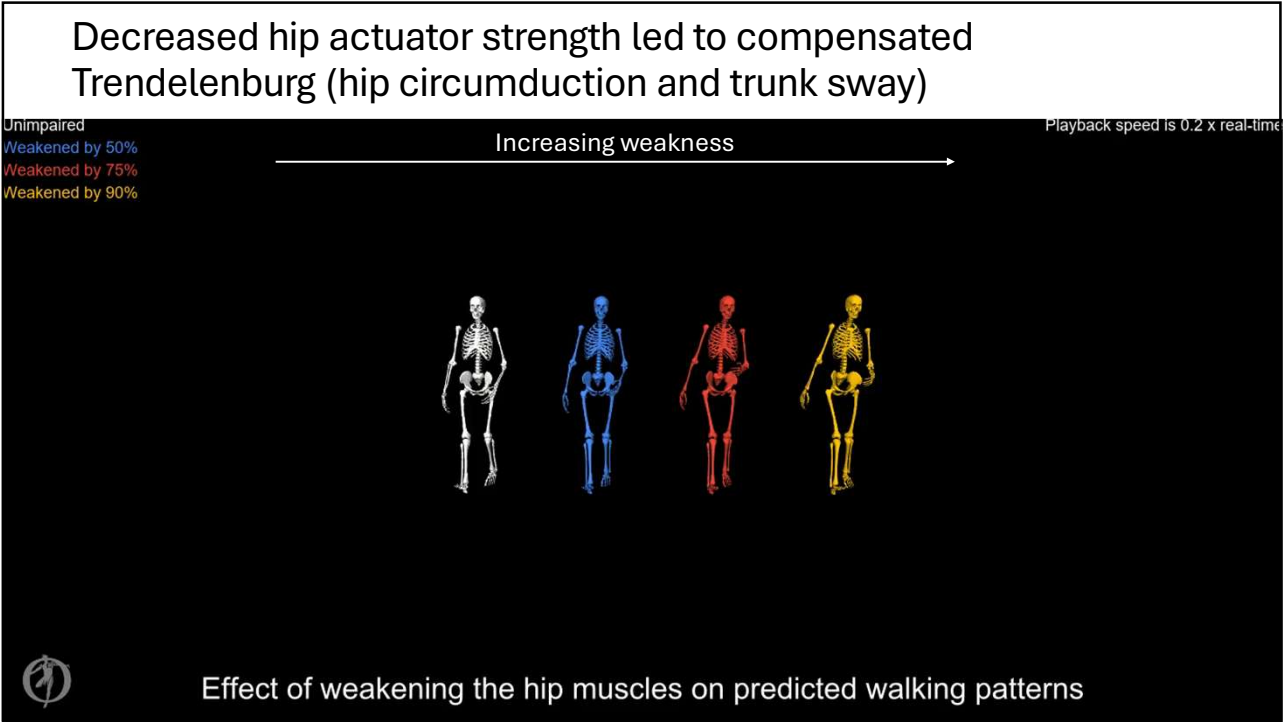
14



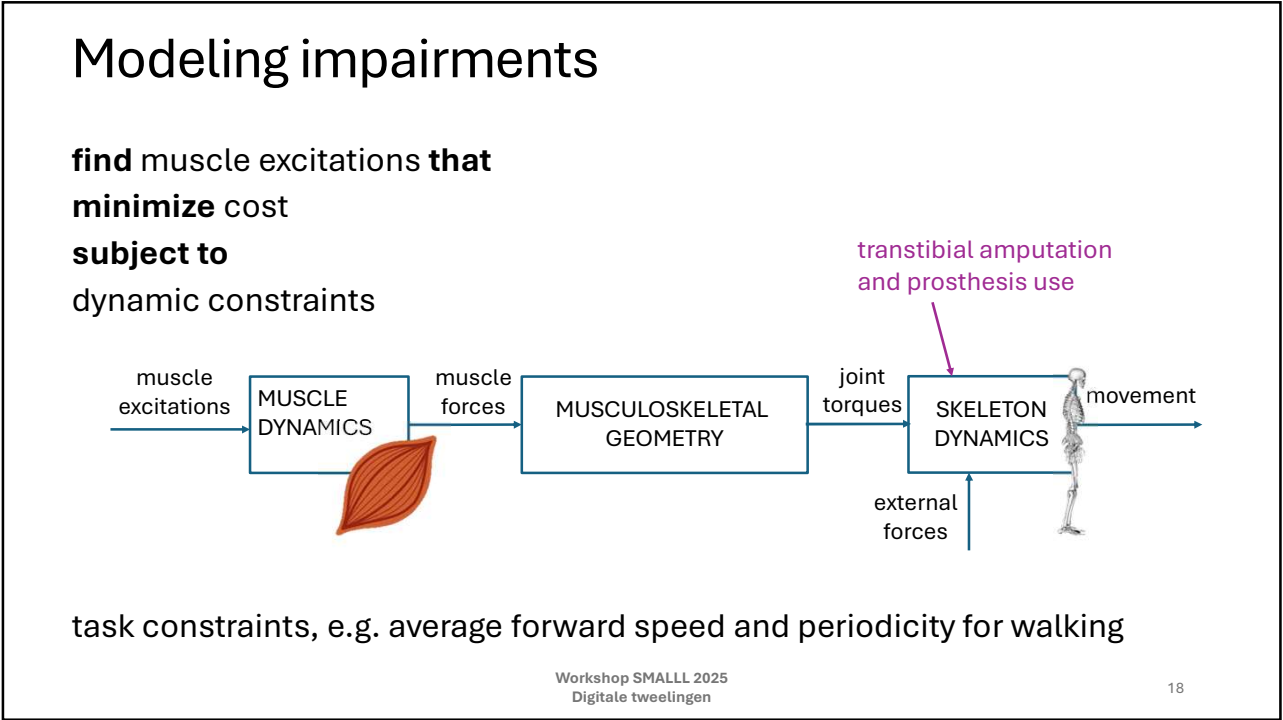
15



16

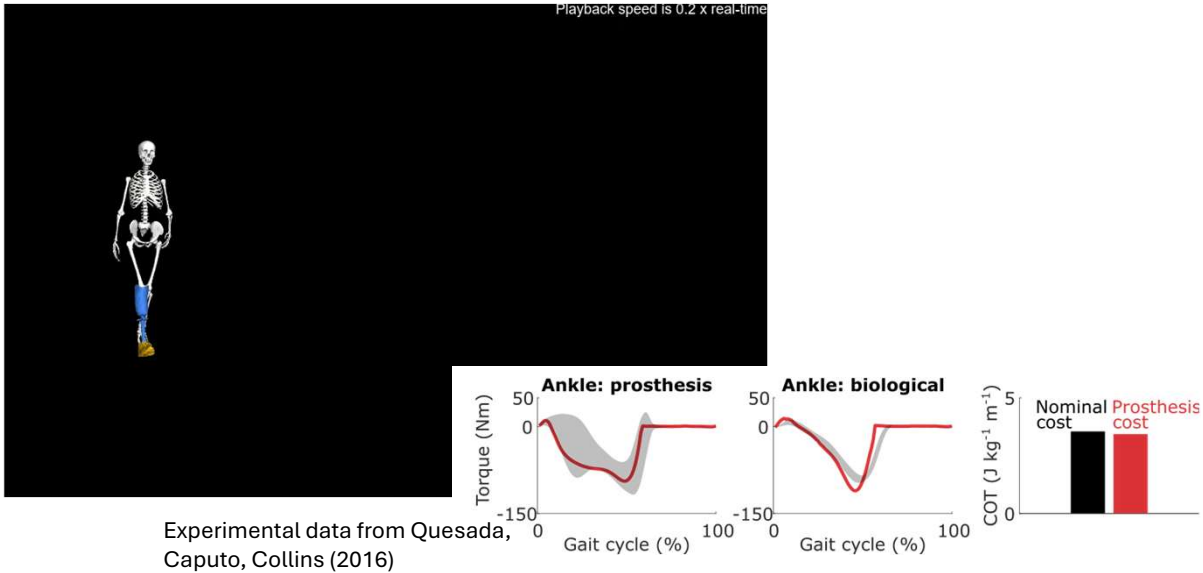


17

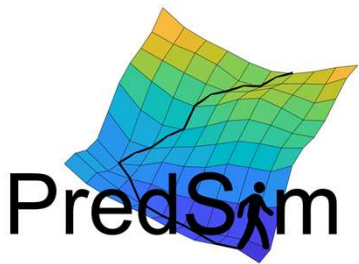


18

Prosthetic use altered the ankle torque
in agreement with experimental data



19



Muscle dynamics
Definition of optimal control problem



Skeleton dynamics using
modified version of OpenSim
that enables algorithmic
differentiation



Musculoskeletal geometry

CasADi for algorithmic
differentiation and formulating
optimization problem
(solved using ipopt)



<https://web.casadi.org/>

20

Why use PredSim software?

PredSim versus OpenSim Moco

	PredSim	OpenSim Moco (Dembia, Bianco et al. <i>PLOS COMP. BIOL.</i> 2020)
Simulation type	Predictive	Predictive, tracking or inverse
Task	Steady-state locomotion	Any task
Discretisation	Direct collocation	Direct collocation
Formulation of dynamics	Implicit	Implicit or explicit
Derivatives	Algorithmic differentiation	Finite differences

AD is faster than FD, and more so for more complex models
(Falisse, Serrancolí et al. *PLOS ONE* 2019)

21

21

Customise and run simulations from a single script

```
% Path to OpenSim model file
osim_path = './Subjects/Falisse_et_al_2022/Falisse_et_al_2022.osim';

% Initialize settings to what was used for this model in the paper
[S] = initializeSettings('Falisse_et_al_2022');

% Path to folder to save the simulation result
S.misc.save_folder = './PredSimResults/Falisse_et_al_2022';

% Select cold-start initial guess
S.solver.IG_selection = 'quasi-random';

% Run simulation
runPredSim(S, osim_path);
```

22

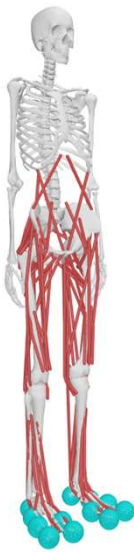
22

OpenSim model file

```
% Path to OpenSim model file
osim_path = './Subjects/Falisse_et_al_2022/Falisse_et_al_2022.osim';
```

Example models:

• Falisse_et_al_2022	31 DoFs	92 muscles
• DHondt_et_al_2024_3seg	31 DoFs	92 muscles
• DHondt_et_al_2024_4seg	33 DoFs	94 muscles
• Gait1018	10 DoFs	18 muscles



23

A few required settings ...

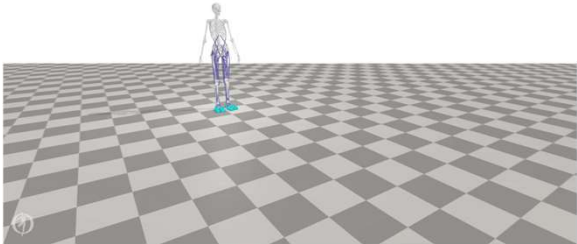
```
% Path to OpenSim model file
osim_path = './Subjects/Falisse_et_al_2022/Falisse_et_al_2022.osim';

% Initialize settings to what was used for this model in the paper
[S] = initializeSettings('Falisse_et_al_2022');

% Path to folder to save the simulation result
S.misc.save_folder = './PredSimResults/Falisse_et_al_2022';

% Select cold-start initial guess
S.solver.IG_selection = 'quasi-random';
```

} important settings: no defaults



- Alternative initial guess: OpenSim motion file
- Experimental data
 - Previous result
 - Custom

24

... and many optional settings

- subject
- orthosis
- metabolicEnergy
- weights
- solver
- bounds
- OpenSimADOptions
- post_processing
- misc (task constraints, file paths, scaling, ...)

→ This workshop

```
% Average forward speed of pelvis
S.misc.forward_velocity = 0.8;

% Periodic and symmetric
S.misc.gaitmotion_type = 'HalfGaitCycle';
    initial kinematics, activations
    =
    mirrored final kinematics, activations
    (except forward position)

% Periodic
S.misc.gaitmotion_type = 'FullGaitCycle';
    initial kinematics, activations
    =
    final kinematics, activations
    (except forward position)
```

“initial” can be any point in the cycle
e.g. 80 – 20% is also half cycle

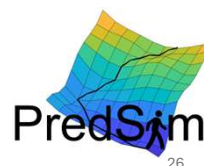
Workshop SMALLL 2025
Digitale tweelingen

25

De casussen van vandaag: Wat kunnen we personaliseren en simuleren?

- Casus 1: Spierzwakte en -stijfheid bij Duchenne spierdystrofie
(Ines Vandekerckhove)
- Casus 2: Chirurgische ingreep bij cerebrale parese
(Ellis Van Can)
- Casus 3: Het effect van een enkel-voet-orthese bij een dropfoot
(Tim van der Zee)

Op basis van 2D musculoskeletale modellen.



Workshop SMALLL 2025
Digitale tweelingen

26

26

KU LEUVEN

Workshop: Predictive simulations

Case 1: Duchenne muscular dystrophy (DMD)

SMALLL congres 2025



DMD

27

Duchenne muscular dystrophy (DMD)

X-linked degenerative disorder

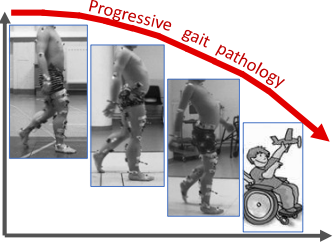
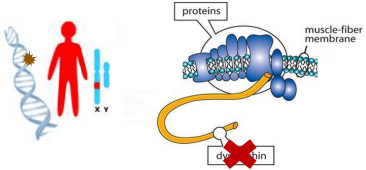
Absence of protein dystrophin

Muscle damage

Muscle impairments

- Progressive muscle weakness
- Progressive muscle stiffness and contractures

Progressive gait pathology



Sussman, 2002, J Am Acad Orhtop Surg; Emery et al., 1991, Neuromuscul Disord; Sutherland et al., 1981, Develop. Med. Child. Neuro.

28

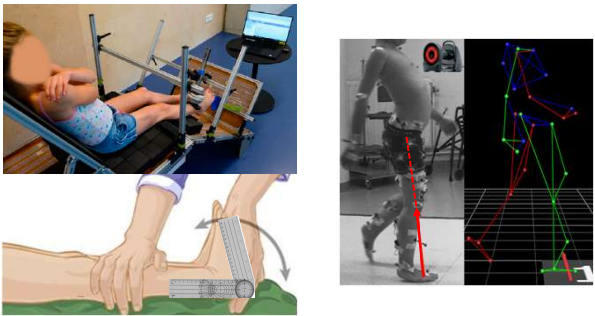
DMD case

- Boy of 15.6 years
- Tiptoeing gait pattern



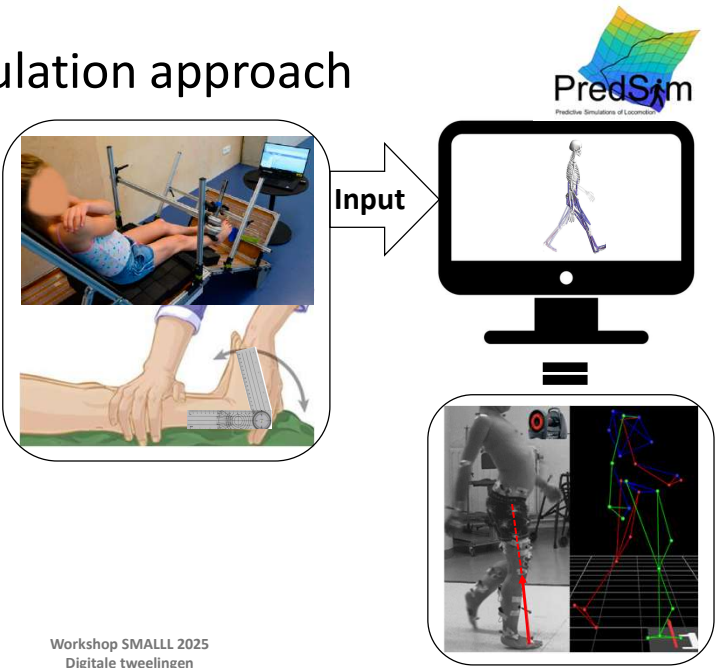
Clinical assessments

- Weakness
 - Fixed dynamometry
- Stiffness and contractures
 - Clinical examination
 - Goniometry
 - Clinical stiffness scale
- Gait pattern
 - 3D-gait analysis

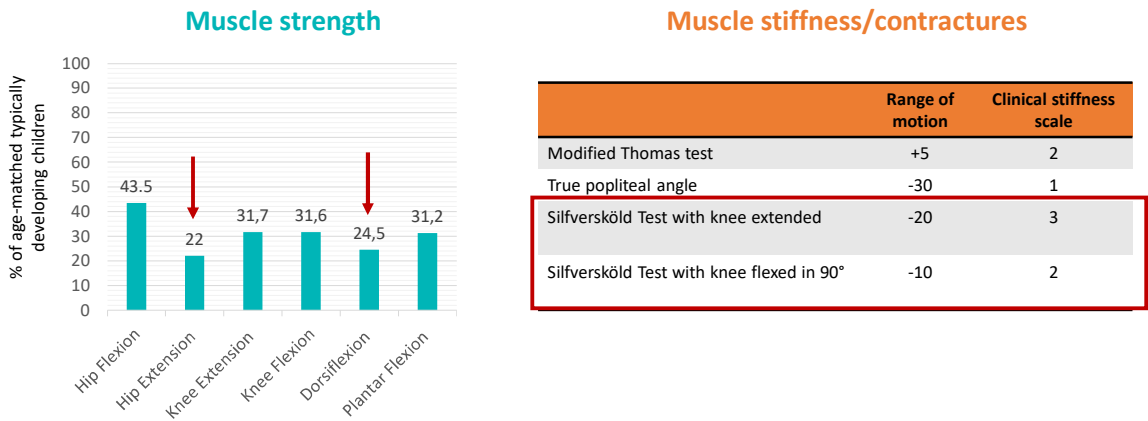


Simulation approach

- Weakness
 - Fixed dynamometry
- Stiffness and contractures
 - Clinical examination
 - Goniometry
 - Clinical stiffness scale
- Gait pattern
 - 3D-gait analysis



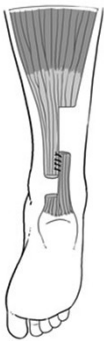
DMD Case: muscle impairments



33

DMD Case: Achilles tendon lengthening

- Common in the past
- Later controversial
 - Some patients lost ambulation directly after Achilles tendon lengthening
- With predictive simulations based on personalized models, you can test whether such an intervention would improve the gait pattern



34



1. Model personalization

- Subject-specific muscle weakness
 - Based on instrumented strength scores
- Subject-specific muscle stiffness
 - Based on passive range of motion and clinical stiffness scale

2. Achilles tendon lengthening simulation

- Increase tendon slack length in the model
- Evaluate whether this intervention is appropriate for this patient

Workshop SMALLL 2025
Digitale tweelingen

35

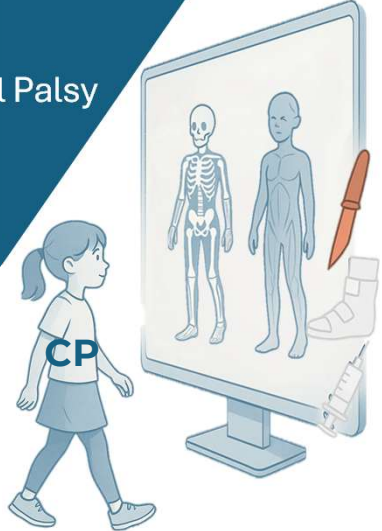
35

KU LEUVEN

Workshop: Predictive simulations

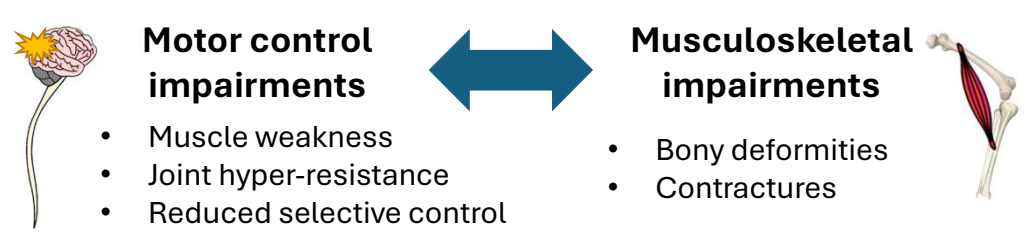
Case 2: Surgical intervention for a child with Cerebral Palsy

SMALLL congres 2025



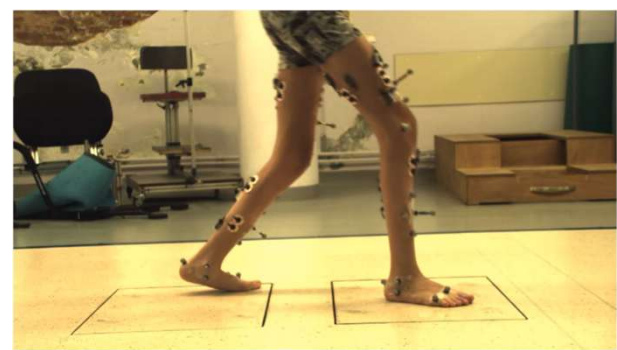
36

Cerebral Palsy (CP)



37

Case CP SMALLL



Clinical Exam				
	L		R	
Muscle weakness				
	FL	Ext	FL	Ext
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

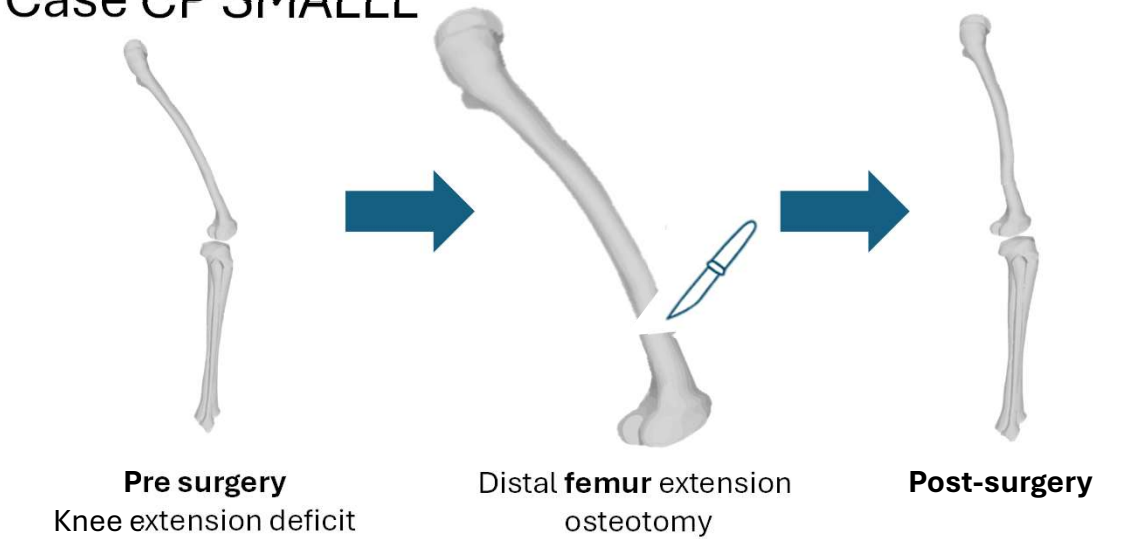


**Single Event
Multilevel Surgery
(SEMS)**

 **Single Event
Multilevel Surgery
(SEMS)**

38

Case CP SMALLL



Pre surgery
Knee extension deficit

Distal **femur** extension osteotomy

Post-surgery

Workshop SMALLL 2025
Digitale tweelingen

39

39

Case CP SMALLL

I. Clinical Exam

	L		R	
	Fl.	Ext.	Fl.	Ext.
Muscle weakness				
Hip				
Knee				
Ankle				
Contractures				
Hamstrings				
Iliopsoas				
Limited Range of Motion				
Knee extension				

II.



III.



PredSim

Workshop SMALLL 2025
Digitale tweelingen

40

40

KU LEUVEN

Workshop: Predictive simulations

Case 3: Simulate the effects of an ankle-foot orthosis on gait patterns in individuals with dropfoot

SMALLL congres 2025



41

Case 3: footdrop / dropfoot (NL: klapvoet)

Common cause: nerve damage

Right Leg: Foot Drop



Damage to peroneal nerve

Inability to lift the foot

Other causes:

- Multiple sclerosis
- Stroke
- Cerebral palsy
- Muscular dystrophy

Workshop SMALLL 2025
Digitale tweelingen

42

42

Case 3: footdrop / dropfoot (NL: klapvoet)

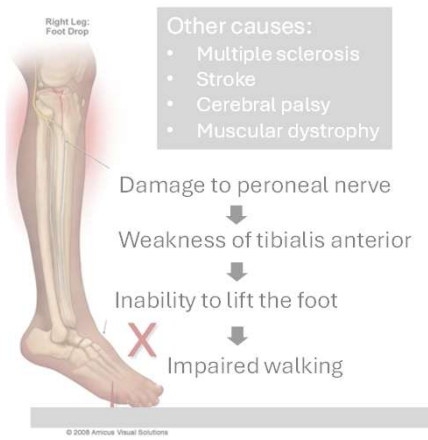
Common cause: nerve damage



43

Case 3: footdrop / dropfoot (NL: klapvoet)

Common cause: nerve damage



Common treatment: ankle-foot orthosis (AFO)



Stewart (2008), *Pract Neurol*, 8: 158-169

44

Case 3: footdrop / dropfoot (NL: klapvoet)

Common cause: nerve damage



Other causes:

- Multiple sclerosis
- Stroke
- Cerebral palsy
- Muscular dystrophy

Damage to peroneal nerve
↓
Weakness of tibialis anterior
↓
Inability to lift the foot
↓
Impaired walking

© 2008 Amicus Visual Solutions

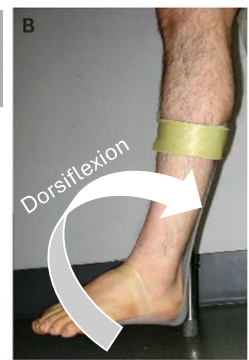
Common treatment: ankle-foot orthosis (AFO)

Device properties:

- Rotational stiffness
- Neutral angle

Unclear which device properties are best

Can we use predictive simulations to inform on AFO design?



Stewart (2008), *Pract Neurol*, 8: 158-169

How to get started with the workshop



- Code sharing
- Documentation (“README”)

How to get started with the workshop

PredSim

<https://github.com/KULeuvenNeuromechanics/PredSim>



Dependencies: Windows, MATLAB, OpenSim, CasADi, Visual Studio, Cmake, Git

- Code sharing
- Documentation ("README")

Step 1: reference simulation with 2D model

PredSim

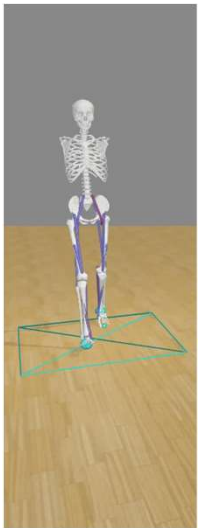
Running a reference 2D simulation with PredSim



Before going to the hands-on tutorials, the user should run a reference 2D simulation with [PredSim](#).

1. Open Matlab
2. Navigate to your `PredSim` folder in Matlab
3. Open the script `main.m` in Matlab by clicking on it
4. During this workshop the user will run predictive simulations with the 2D model instead of the default 3D model. Therefore, adjust the following lines in `main.m`:
 - Line 20 - `[S] = initializeSettings('gait1018');`
 - Line 25 - `S.subject.name = 'gait1018';`
5. Click on the green 'Run' button

Step 1: reference simulation with 2D model



Visualizing your simulation results in OpenSim

To visualize the simulation in OpenSim:

- 1. Open OpenSim
- 2. Click on 'File > Open Model...' and navigate to the 2D model `PredSim/Subjects/gait1018/gait1018.osim`
- 3. Click on 'File > Load Motion...' and navigate to the mot file of your simulation `PredSimResults/gait1018/gait1018_v1.mot`
- 4. Click on the 'Play forward' button to see the motion. You may also adapt the playback speed.

How to get started with the workshop

PredSim



<https://github.com/KULeuvenNeuromechanics/PredSim>



- Code sharing
- Documentation (“README”)

Dependencies: Windows, MATLAB, OpenSim, CasADi, Visual Studio, Cmake, Git

Workshop



<https://github.com/KULeuvenNeuromechanics/PredSim-workshop-smalll-2025>



Dependencies: Windows, MATLAB, OpenSim, CasADi, Visual Studio, Cmake, Git

How to get started with the workshop

Adding PredSim-workshop-smalll-2025 to path

Before starting one of the three cases, make sure that this repository is added to your Matlab path.

1. Either download or clone [the current repository](#)
2. Open Matlab
3. Navigate to the `PredSim-workshop-smalll-2025` folder
4. Open the script called `set_up_paths.m`
5. Click on the green 'Run' button

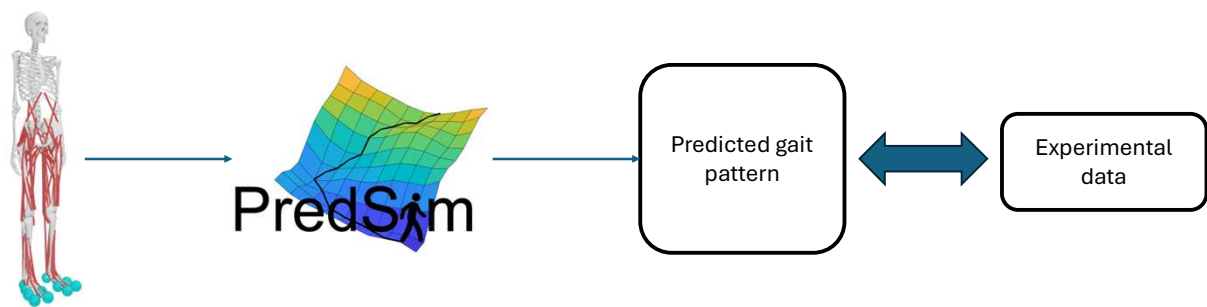
Workshop



<https://github.com/KULeuvenNeuromechanics/PredSim-workshop-smalll-2025>

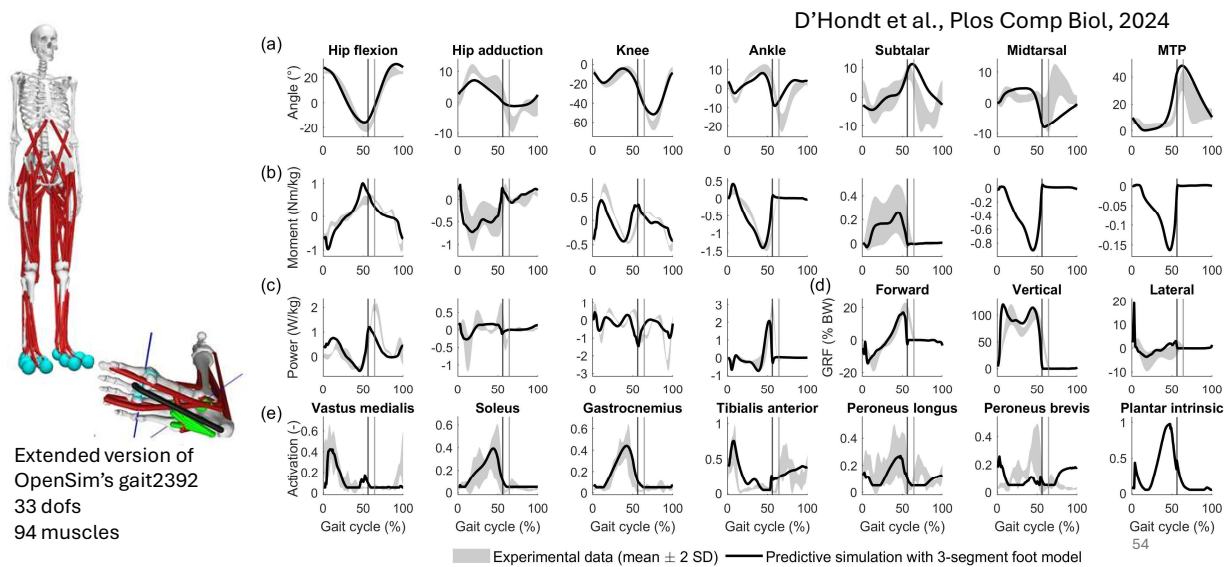
Predictive simulations in 3D

Validation



53

Simulations based on DHondt_et_al_2024 capture walking pattern of unimpaired adult

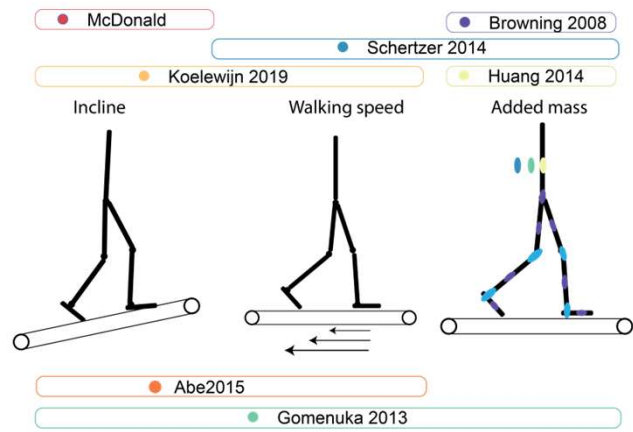


54

Do optimal control simulations predict gait mechanics and energetics across walking conditions?



Maarten Afschrift

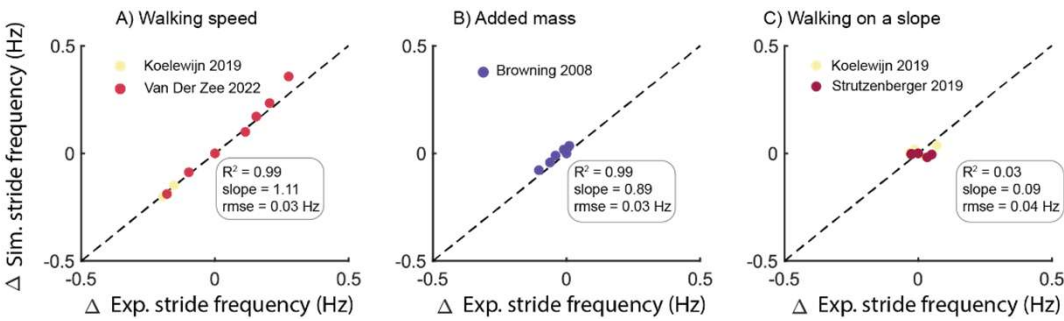


Falisse et al., *Plos One*, 2022; Afschrift et al., *bioRxiv*, 2025

55

55

Simulations capture stride frequency across conditions



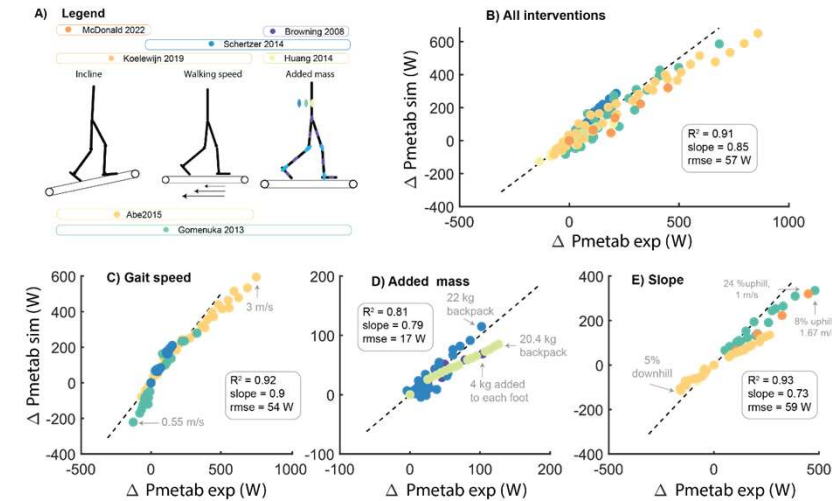
Afschrift et al., *bioRxiv*

Workshop SMALLL 2025
Digitale tweelingen

56

56

Optimal control simulations capture main trends but not fine details of healthy gait energetics



Overall, predictions of mechanics/ activations are better than predictions of metabolic energy.

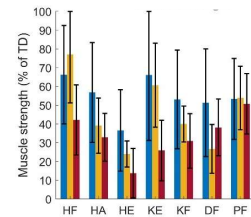
57

Contractures in Duchenne Muscular Dystrophy
- Should we treat them or are they a useful compensation strategy?

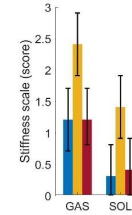


Ines Vandekerckhove

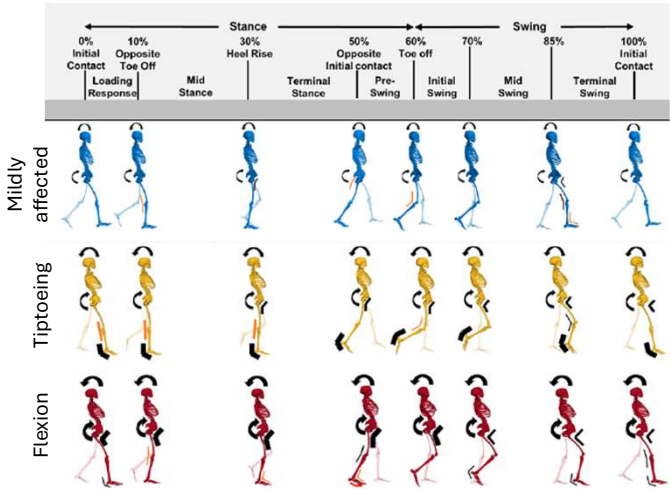
WEAKNESS



CONTRACTURES



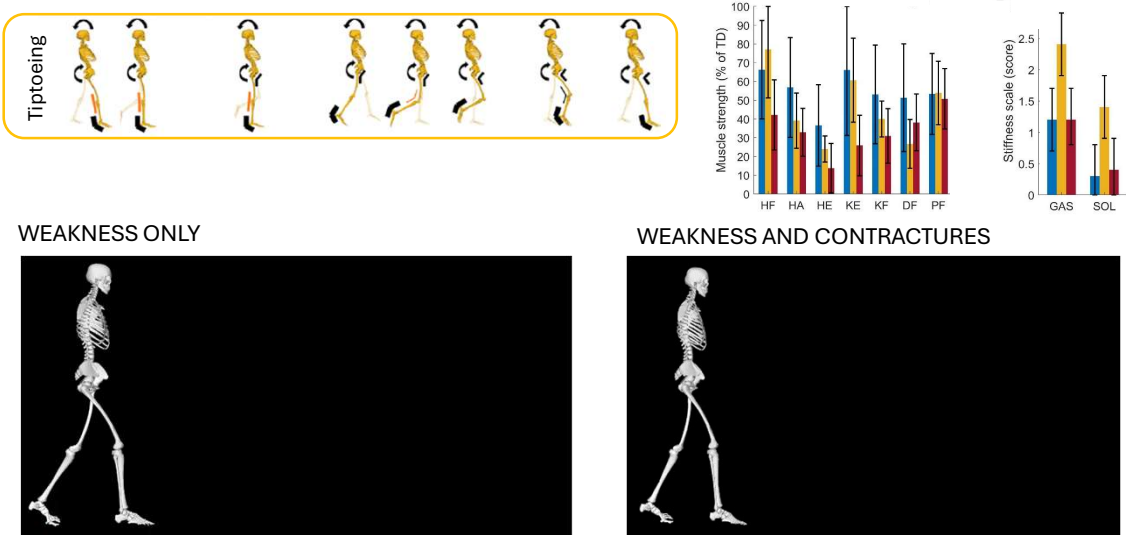
Walking patterns based on a large longitudinal (30 boys – 137 measurements)



Vandekerckhove et al., Scientific Reports, 2024

58

Contractures in Duchenne Muscular Dystrophy
- Should we treat them or are they a useful compensation strategy?



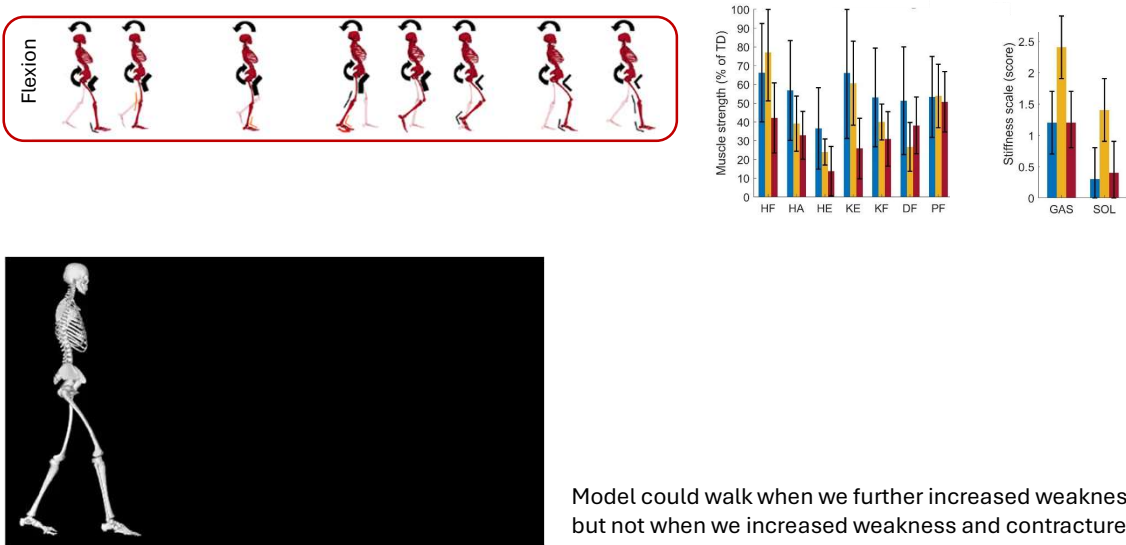
Tiptoeing in the absence of contractures!

Workshop SMALLL 2025
Digitale tweelingen

Vandekerckhove et al., JNER, 2025

59

Contractures in Duchenne Muscular Dystrophy
- Should we treat them or are they a useful compensation strategy?

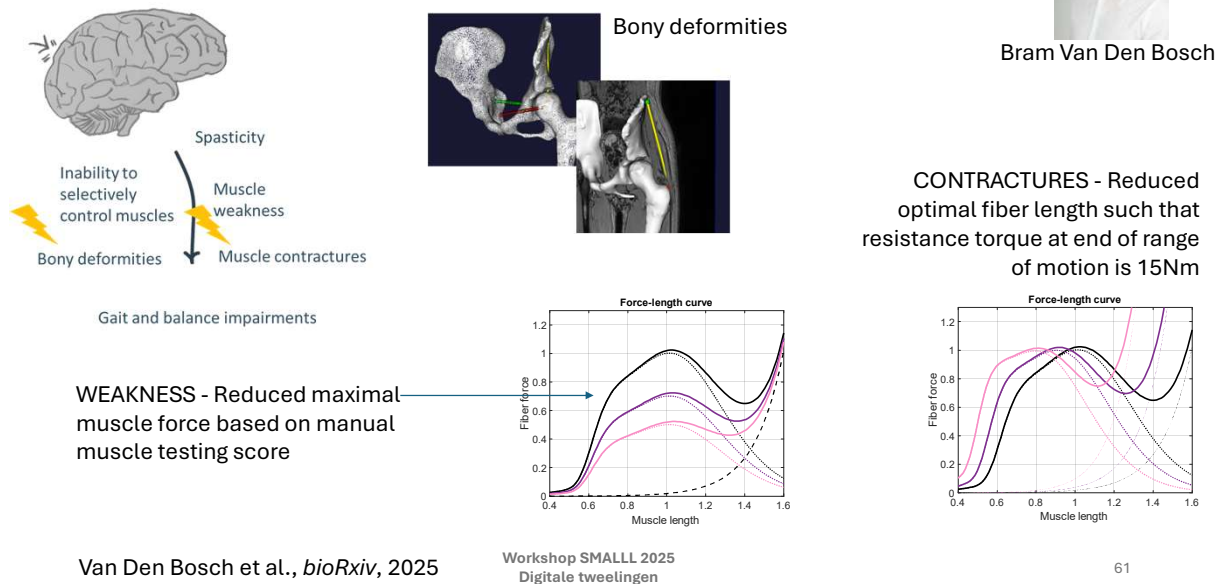


Workshop SMALLL 2025
Digitale tweelingen

Vandekerckhove et al., JNER, 2025

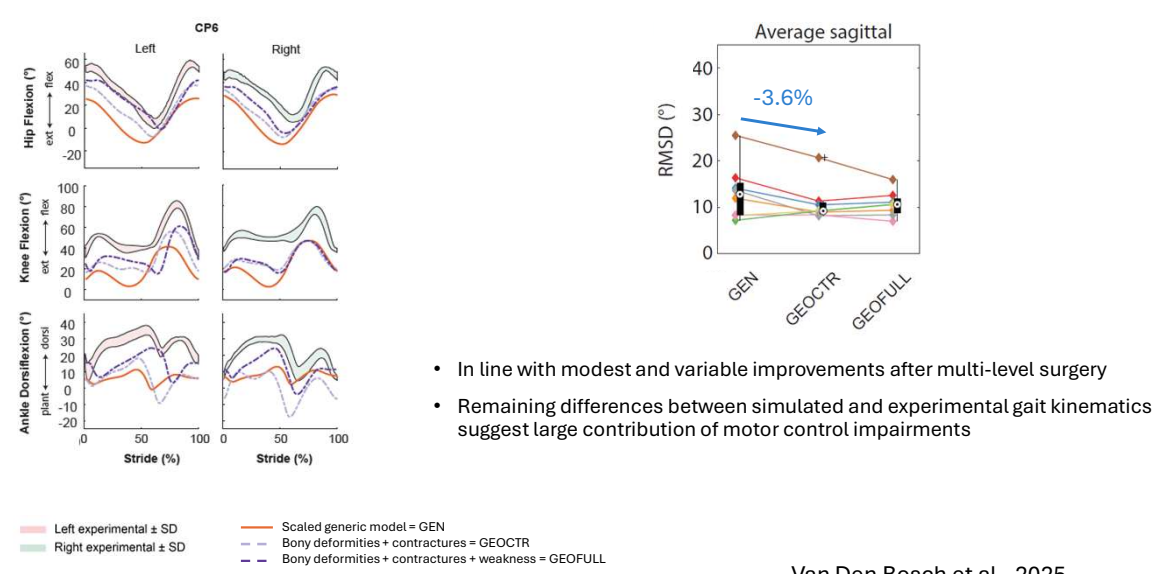
60

What is the contribution of musculoskeletal deficits to alterations in gait kinematics in cerebral palsy?



61

Modest and variable contributions of musculoskeletal impairments to alterations in gait kinematics in cerebral palsy

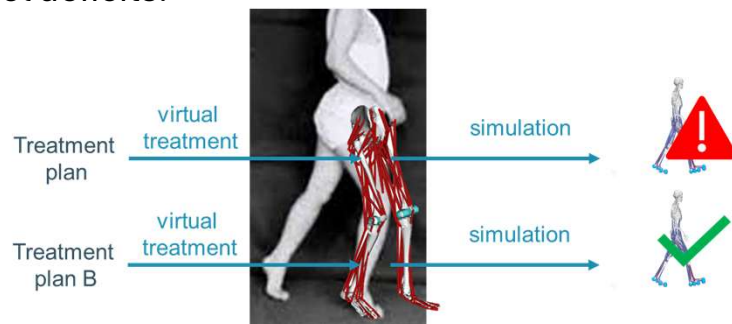


62

We still need to demonstrate benefits of using predictive simulations for treatment selection and design

Challenges

- Improve accuracy of simulations.
- Model motor control deficits.



63

63

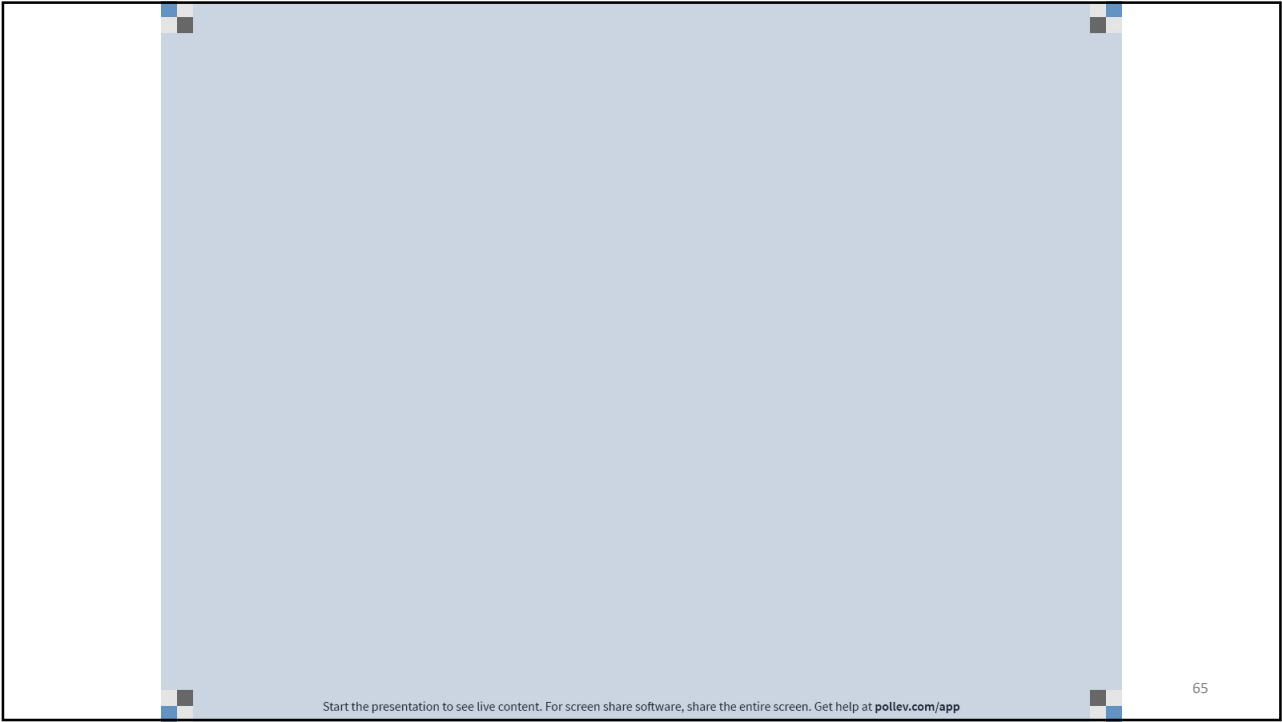
Kansen en obstakels voor de klinische implementatie van predictieve simulaties



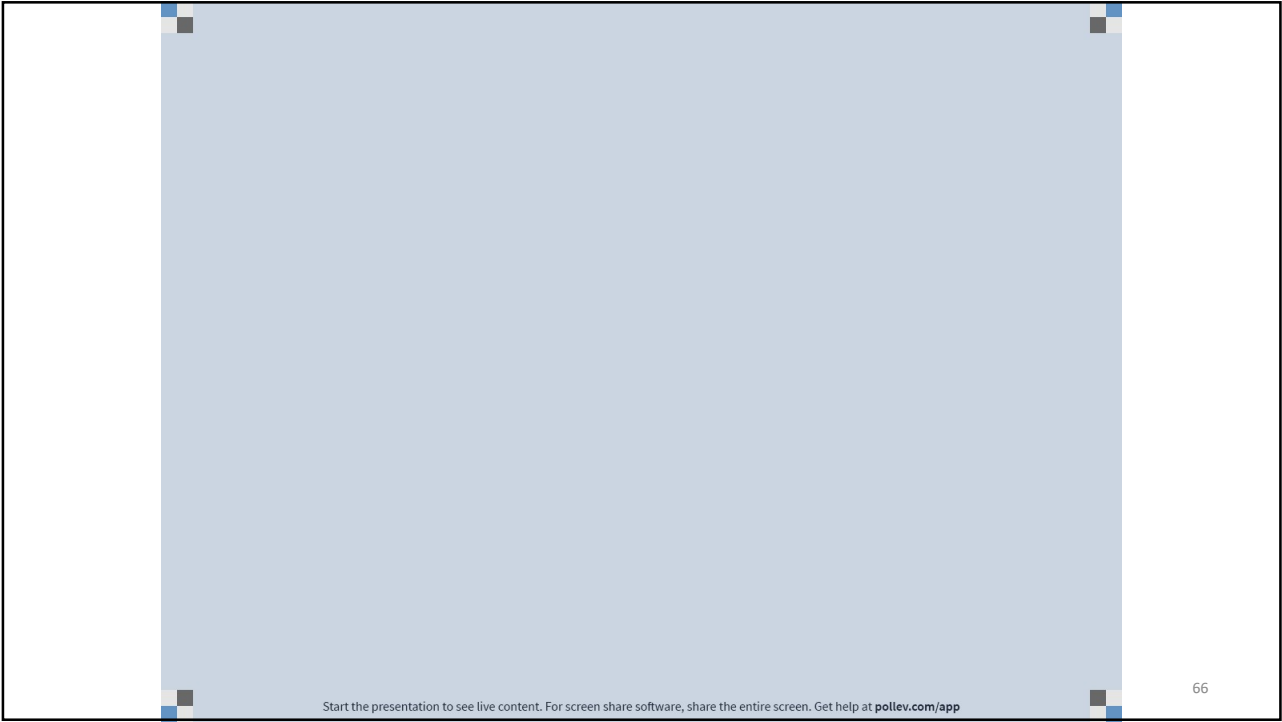
PollEv.com/tombuurke748

64

64



65



66

Het doel van vandaag

Nu...

- kent u de onderliggende theorie van musculoskeletaal modelleren en predictieve simulaties.
- kunt u predictieve simulaties met 2D modellen in *PredSim* uitvoeren.
- kunt u de informatie en bronnen voor 3D simulaties vinden.
- kent u kansen en obstakels voor de klinische implementatie van predictieve simulaties.



Workshop SMALLL 2025
Digitale tweelingen

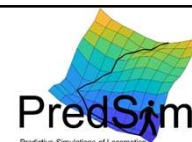
67

67

Digitale tweelingen

Predictieve simulaties van wandelen voor de klinische gangbeeldanalyse

Tom Buurke, Friedl De Groote, Tim van der Zee, Ellis Van Can, Ines Vandekerckhove
Workshop SMALLL Congres 2025, Erasmus MC, Rotterdam, NL



umcg:



KU LEUVEN

68

68